

Employee Attendance and Productivity Tracker

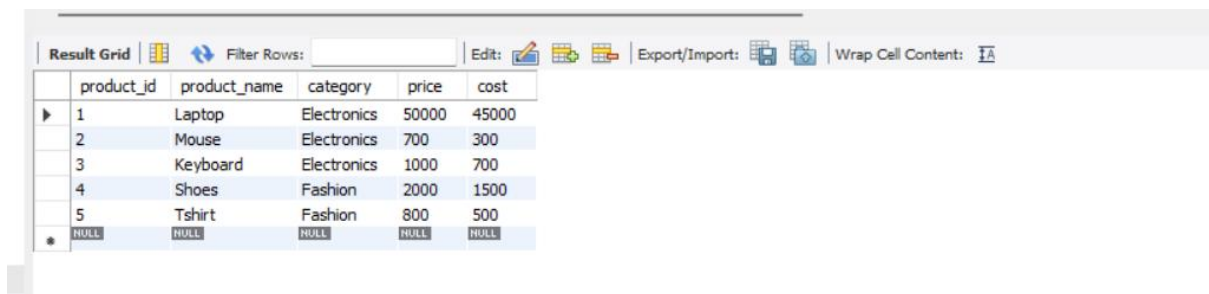
- By Santhosh Kumar V S

MYSQL IMPLEMENTATION:

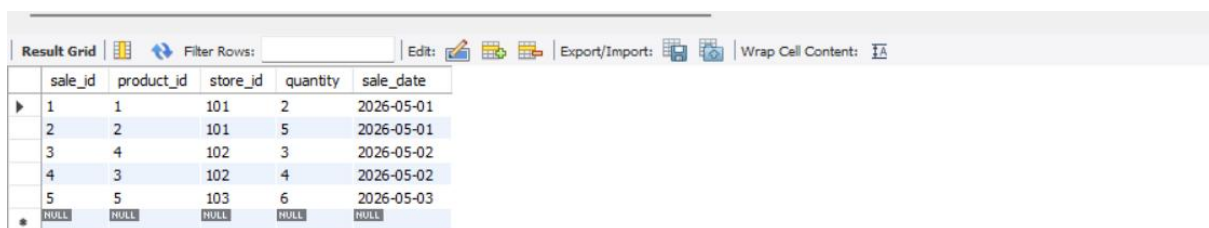
Creating Database and Tables:

```
1 • create database retail_project;
2
3 use retail_project;
4
5
```

Inserted Product and Sales Data:

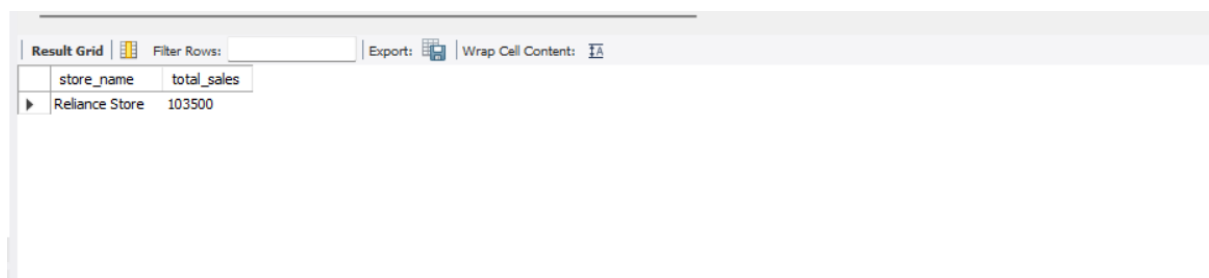


	product_id	product_name	category	price	cost
▶	1	Laptop	Electronics	50000	45000
	2	Mouse	Electronics	700	300
	3	Keyboard	Electronics	1000	700
	4	Shoes	Fashion	2000	1500
	5	Tshirt	Fashion	800	500
*	NULL	NULL	NULL	NULL	NULL



	sale_id	product_id	store_id	quantity	sale_date
▶	1	1	101	2	2026-05-01
	2	2	101	5	2026-05-01
	3	4	102	3	2026-05-02
	4	3	102	4	2026-05-02
	5	5	103	6	2026-05-03
*	NULL	NULL	NULL	NULL	NULL

Stored Procedure Execution:



	store_name	total_sales
▶	Reliance Store	103500

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MONGODB IMPLEMENTATION:

MongoDB Feedback Data Insertion:

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a059d8037b29f4e4c8d12cb'),
    '1': ObjectId('6a059d8037b29f4e4c8d12cc'),
    '2': ObjectId('6a059d8037b29f4e4c8d12cd')
  }
}
```

MongoDB Collection Data:

```
> db.campaign_feedback.find().pretty()
< {
  _id: ObjectId('6a059d8037b29f4e4c8d12cb'),
  customer_name: 'Rahul',
  product: 'Laptop',
  feedback: 'Good product',
  rating: 4
}
{
  _id: ObjectId('6a059d8037b29f4e4c8d12cc'),
  customer_name: 'Anjali',
  product: 'Shoes',
  feedback: 'Nice quality',
  rating: 5
}
{
  _id: ObjectId('6a059d8037b29f4e4c8d12cd'),
  customer_name: 'Kiran',
  product: 'Mouse',
  feedback: 'Average',
  rating: 3
}
```

MongoDB Index Creation:

```
> db.campaign_feedback.createIndex(
  {
    product:1
  }
)
< product_1
```

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PYTHON IMPLEMENTATION:

Python Script Execution:

```
SANTHOSH SELVA@DESKTOP-G5J7BLI MINGW64 /e/Employee Attendance and Productivity Tracker
$ python python/sales_analysis.py
sale_id product_id store_id quantity sale_date
0 1 1 101 2 2026-05-01
1 2 2 101 5 2026-05-01
2 3 4 102 3 2026-05-02
3 4 3 102 4 2026-05-02
4 5 5 103 6 2026-05-03

sale_id product_id store_id quantity sale_date ... cost store_name city revenue profit
0 1 1 101 2 2026-05-01 ... 45000 Reliance Store Mumbai 100000 10000
1 2 2 101 5 2026-05-01 ... 300 Reliance Store Mumbai 2500 1000
2 3 4 102 3 2026-05-02 ... 1500 Dmart Pune 6000 1500
3 4 3 102 4 2026-05-02 ... 700 Dmart Pune 4000 1200
4 5 5 103 6 2026-05-03 ... 500 Big Bazaar Delhi 4800 1800
5 6 1 103 1 2026-05-03 ... 45000 Big Bazaar Delhi 50000 5000

[6 rows x 13 columns]
revenue profit
store_name
Big Bazaar 54800 6800
Dmart 10000 2700
Reliance Store 102500 11000

sale_id product_id store_id quantity sale_date ... cost store_name city revenue profit
0 1 1 101 2 2026-05-01 ... 45000 Reliance Store Mumbai 100000 10000
2 3 4 102 3 2026-05-02 ... 1500 Dmart Pune 6000 1500
5 6 1 103 1 2026-05-03 ... 45000 Big Bazaar Delhi 50000 5000

[3 rows x 13 columns]
file saved
```

Generated Output CSV File:

```
output > final_sales.csv > data
1 sale_id,product_id,store_id,quantity,sale_date,product_name,category,price,cost,store_name,city,revenue,profit
2 1,1,101,2,2026-05-01,Laptop,Electronics,50000,45000,Reliance Store,Mumbai,100000,10000
3 2,2,101,5,2026-05-01,Mouse,Electronics,500,300,Reliance Store,Mumbai,2500,1000
4 3,4,102,3,2026-05-02,Shoes,Fashion,2000,1500,Dmart,Pune,6000,1500
5 4,3,102,4,2026-05-02,Keyboard,Electronics,1000,700,Dmart,Pune,4000,1200
6 5,5,103,6,2026-05-03,Tshirt,Fashion,800,500,Big Bazaar,Delhi,4800,1800
7 6,1,103,1,2026-05-03,Laptop,Electronics,50000,45000,Big Bazaar,Delhi,50000,5000
```

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PYSPARK IMPLEMENTATION:

PySpark Store-Level Analysis:

```
cols = ["store_id", "quantity"]
df = spark.createDataFrame(data, cols)
print("sales data")
df.show()
```

```
sales data
+-----+-----+
|store_id|quantity|
+-----+-----+
|    101|        2|
|    101|        5|
|    102|        3|
|    102|        4|
|    103|        6|
+-----+-----+
```

```
total = df.groupBy("store_id").sum("quantity")
print("store total quantity")
total.show()
```

```
store total quantity
+-----+-----+
|store_id|sum(quantity)|
+-----+-----+
|    101|            7|
|    103|            6|
|    102|            7|
+-----+-----+
```

```
▶ low = df.filter(df.quantity < 3)
print("low sales")
low.show()
```

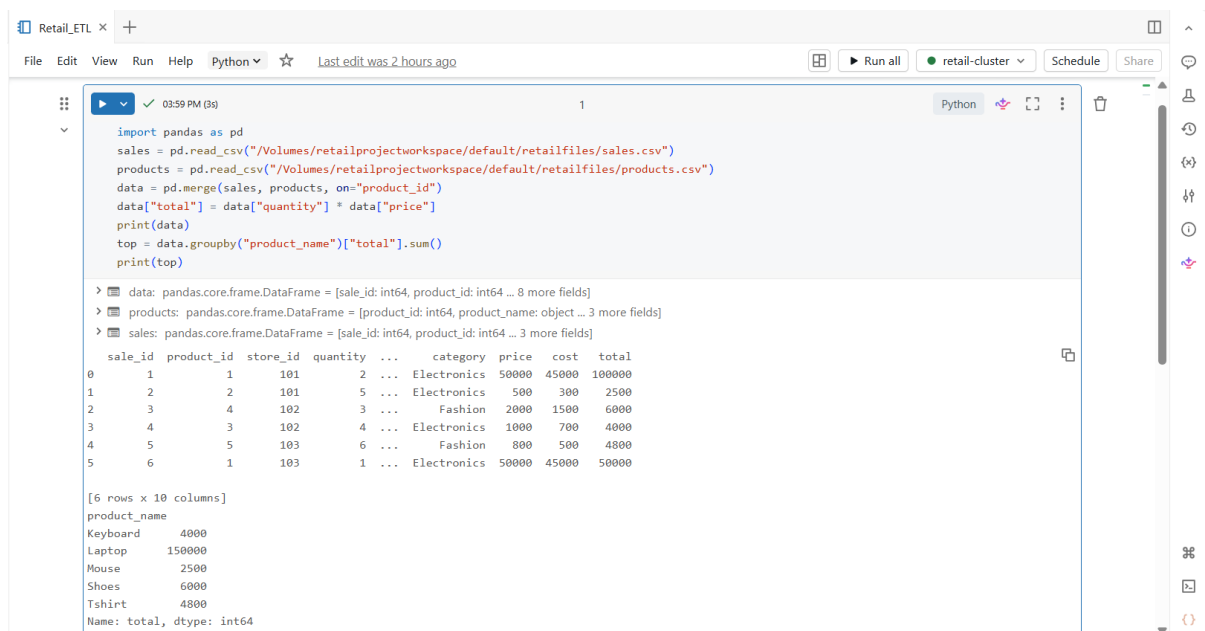
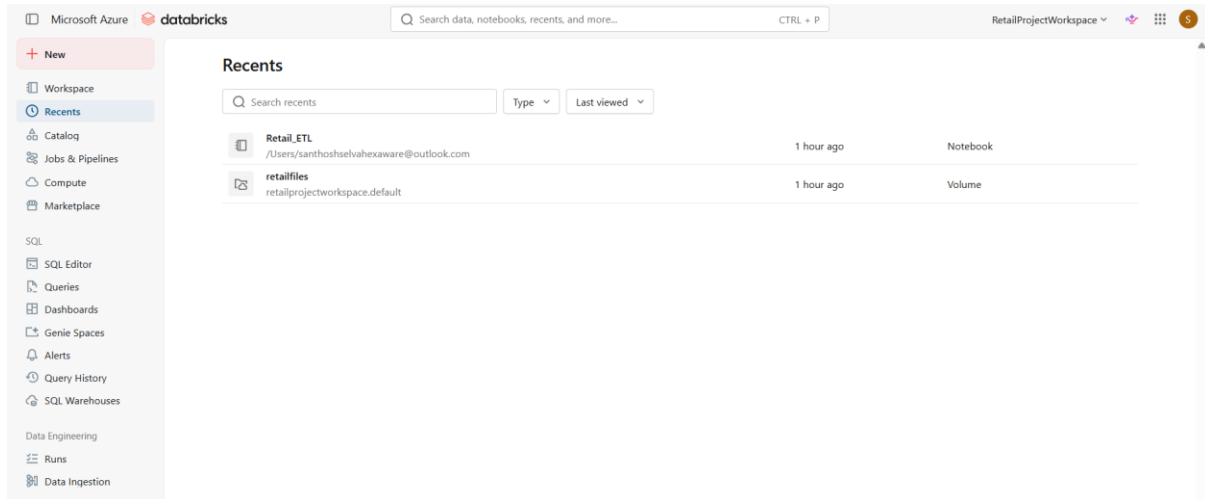
```
... low sales
+-----+-----+
|store_id|quantity|
+-----+-----+
|    101|        2|
+-----+-----+
```

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AZURE DATABRICKS IMPLEMENTATION:

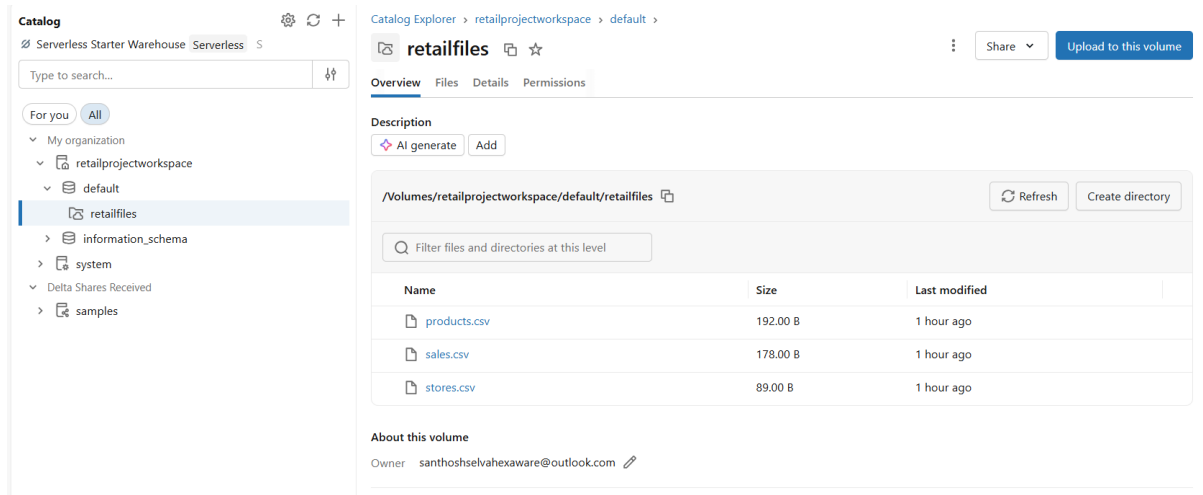
AZURE DATABRICKS ETL PROCESS:



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Uploaded CSV Files in Databricks:



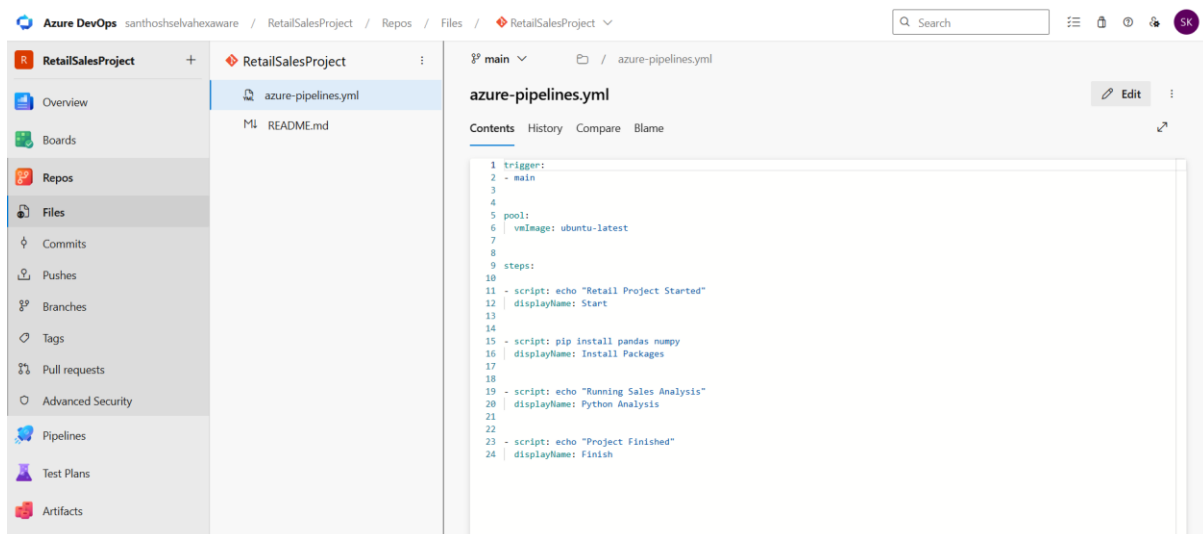
The screenshot shows the Databricks Catalog Explorer interface. On the left, there's a sidebar with a search bar and a list of volumes under 'My organization'. The main area displays the 'retailfiles' volume, which contains three CSV files: 'products.csv' (192.00 B), 'sales.csv' (178.00 B), and 'stores.csv' (89.00 B). The files were last modified 1 hour ago. The interface also includes a 'Description' section with an 'AI generate' button and a 'Refresh' button for the volume.

Name	Size	Last modified
products.csv	192.00 B	1 hour ago
sales.csv	178.00 B	1 hour ago
stores.csv	89.00 B	1 hour ago

AZURE DEVOPS IMPLEMENTATION:

AZURE DEVOPS PIPELINE:

Azure DevOps Pipeline Configuration



The screenshot shows the Azure DevOps interface for configuring a pipeline. The pipeline is named 'azure-pipelines.yml' and is located in the 'RetailSalesProject' repository. The pipeline configuration includes a trigger for the 'main' branch and a series of steps: 'main', 'pool', 'vmImage: ubuntu-latest', and 'steps' which includes 'script: echo \"Retail Project Started\"', 'script: pip install pandas numpy', 'script: echo \"Running Sales Analysis\"', and 'script: echo \"Project Finished\"'.

```
1 trigger:
2   - main
3
4
5 pool:
6   vmImage: ubuntu-latest
7
8 steps:
9
10  - script: echo "Retail Project Started"
11    displayName: Start
12
13
14  - script: pip install pandas numpy
15    displayName: Install Packages
16
17
18  - script: echo "Running Sales Analysis"
19    displayName: Python Analysis
20
21
22  - script: echo "Project Finished"
23    displayName: Finish
24
```

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Azure DevOps Pipeline Execution:

This screenshot shows the 'RetailSalesProject' overview in Azure DevOps. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines (selected), Pipelines (sub-link), Environments, Library, Test Plans, and Artifacts. The main area displays the 'Runs' tab for the selected pipeline. A single run is listed with a green checkmark, indicating success. The run details include the description '#20260514.1 - Set up CI with Azure Pipelines', the repository 'Individual CI for main', and the commit 'd802f740'. The run was completed 2 hours ago and took 59 seconds.

This screenshot shows the details of a specific pipeline run (#20260514.1) in Azure DevOps. The left sidebar is the same as the previous screenshot. The main area shows the 'Summary' tab. A message at the top states: 'This run is being retained as one of 3 recent runs by main (Branch)'. Below this, the run is identified as 'Individual CI by Santhosh Kumar' initiated 'Today at 4:11 PM' and lasting '59s'. It is linked to the 'RetailSalesProject' repository, 'main' branch, and commit 'd802f740'. The 'Stages' section shows one stage completed in 55 seconds, which contains a single job completed in 12 seconds. A 'Rerun Stage' button is visible. The right sidebar contains links for 'View retention leases', 'View 2 changes', 'Parameters', and 'Tests and coverage'.

This screenshot shows the details of a specific job within the pipeline run (#20260514.1) in Azure DevOps. The left sidebar is the same as the previous screenshots. The main area shows the 'Jobs' tab for the selected stage. A single job is listed with a green checkmark, indicating success, and a duration of 12 seconds. The right sidebar contains a 'View raw log' button. The job log is displayed in a dark-themed window with the following content:

```
1 Pool: Azure Pipelines
2 Image: ubuntu-latest
3 Agent: Azure Pipelines 1
4 Started: Today at 4:12 PM
5 Duration: 12s
6
7 > Job preparation parameters
```